

Generators and Relations

Df a subset $S \subseteq G$ generates G iff the smallest subgrp containing S is G itself

Prob S generates G if and only if the hom F_S to G induced by the inclusion S to G is surjective

[usually, F_S is much larger than G]

note: if S generates G , then

$$F_S/K \cong G, \quad \text{where } K = \ker(F_S \text{ to } G)$$

Df a subset $R \subseteq F_S$ is a set of relations for G wrt S iff the smallest normal subgrp of F_S containing R is K

in this case, we write $G = \langle s \in S \mid r \in R \rangle$
and say G is generated by S modulo R

Ex up to iso, a unique group of size 2:

$$G = \{e, s\} \quad \text{s.t.} \quad \begin{aligned} ee &= ss = e \\ se &= es = s \end{aligned}$$

take $S = \{s\}$

then $\ker(F_S \text{ to } G) = \{\text{even powers of } s\}$
altogether $G = \langle s \mid s^2 \rangle$

Rem any G has a “silly” generating set: $[?]$
take $S = G$ itself
[usually we prefer a smaller S]

Ex take $G = \mathbb{Z}$
what is a one-elt generating set?
 $S = \{1\}$ works [but:]
 $S = \{-1\}$ also works

a two-elt generating set without ± 1 ? $S = \{2, 3\}$